Status	Finished
Started	Wednesday, 21 October 2026, 8:38 PM
Completed	Wednesday, 21 October 2026, 8:43 PM
Duration	4 mins 19 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The lower middle element is chosen as the middle element ✗
- b. The search stops
- c. The higher middle element is chosen as the middle element
- d. The middle element is chosen randomly

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is very small
- b. When the list contains duplicate elements
- c. When the list is sorted ✓
- d. When the list is unsorted

Question **3**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search can find multiple instances of the target element
- b. Binary search has a lower time complexity on large, sorted lists ✓
- c. Binary search works on unsorted lists
- d. Binary search does not require dividing the list

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist

b. Number values should only be present

Number values should only be present

c. There must be a mechanism to delete and/or insert elements in the list ✓

d. The list must be sorted

Question **5**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It can be applied only if the list is sorted ✓

b. It works on unsorted lists

c. It compares elements sequentially

d. It always starts from the beginning of the list

Question **6**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Divide the list into two equal parts

b. Compare the target element with the middle element in the list ✓

c. Compare the target element with the first element in the list

d. Sort the list

Question **7**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O(1) ✓
- b. O(log n)
- c. O🥐
- d. O(n log n)

Question **8**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 89 and 94
- b. 89 and 99
- c. 90 and 99 ✓
- d. 90 and 94

Question **9**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Hash
- b. Binary ✓
- c. Both (1) & (3)
- d. Linear

Question **10**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the list is sorted
- b. When the element is found in the list ✓
- c. When the list contains duplicate elements
- d. When the search algorithm finishes

Question **11**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O(1)
- b. O👎 ✓
- c. O(log n)
- d. O(n log n)

Question **12**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The first element is returned
- b. An error is raised
- c. The last element is returned
- d. The search is considered unsuccessful ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and unsorted ✓
- b. When the list is small and sorted
- c. When the list is large and unsorted
- d. When the list is large and sorted

Question **14**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. log n
c. n ✓
d. n-1

Question **15**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the left sublist
b. At the beginning of the list
c. In the middle of the list
d. In the right sublist ✓